LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Quick Revision: Our Environment

Ecosystem

- **Ecosystem** = all living organisms in an area + the non-living environment. **Biotic:** plants, animals, microbes. **Abiotic:** temperature, rainfall, wind, soil, minerals.
- Natural ecosystems: forests, ponds, lakes. Artificial ecosystems: gardens, crop fields, aquariums.
- **Producers** (green plants) → **consumers** (herbivores, carnivores, omnivores, parasites) → **decomposers** (bacteria, fungi) return nutrients to the soil.

Food chains & energy flow

- Each step of a food chain is a **trophic level**: grass (producer) → grasshopper (primary consumer) → frog (secondary) → snake (tertiary) → eagle.
- Green plants capture about **1%** of the sunlight falling on their leaves.
- **10 per cent law:** only about 10% of the energy is passed to the next level. The rest is used for life processes or lost as heat.
- So food chains have only **3–4 levels**, and there are fewer organisms at higher levels.
- Energy flow is **unidirectional**: it never goes back to the previous level. Food chains interconnect to form a **food web**.

Biological magnification: non-biodegradable chemicals (e.g. pesticides) are not broken down, so they build up at each trophic level. Their concentration is highest at the **top** of the food chain, including in humans.

Ozone layer

- Ozone (O_3) in the upper atmosphere shields the Earth from harmful **UV radiation** (which causes skin cancer).
- Formation: $O_2 \rightarrow O + O$ (UV splits oxygen); then $O + O_2 \rightarrow O_3$
- Depleted by **CFCs** (used in refrigerants and fire extinguishers). In 1987, UNEP got countries to agree to freeze CFC production at 1986 levels. Now CFC-free refrigerators are made.

Managing garbage

Biodegradable	Non-biodegradable
Broken down by microbes	Not broken down; stays for a long time
Peels, paper, wood, cotton, food waste	Plastic, polythene, glass, DDT, some metals

- Reduce waste: segregate wet and dry, compost kitchen waste, carry cloth bags, avoid single-use plastic, reuse, recycle.
- Even biodegradable options have costs: kulhads (clay cups) on trains would use up fertile top soil, so paper cups are used.

Quick check

1. If 20,000 J is available to the producers, how much reaches the secondary consumers?

2. Why is energy flow in an ecosystem called unidirectional?